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A low-power 18-bit sigma-delta digital-to-analog converter with low-temperature-drift reference

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ABSTRACT

An 18-bit Σ-Δ digital-to-analog converter (DAC) with precision-trimmed bandgap reference is proposed for industrial transmitters. A bandgap reference with an 8-bit trimming DAC and a high-order compensation circuit for reducing the temperature coefficient (TC) affected by process and voltage variation was utilized, which effectively guarantees the robustness of the Σ-Δ DAC. Compared with the conventional Σ-Δ DAC, the proposed DAC replaces the on-chip digital interpolation filter (IF) by providing oversampled input digital codes, and a low-pass configurable off-chip passive RC filter is utilized for power reduction. A quantization noise randomization scheme was employed by adding pseudo-random sequences in the Σ-Δ modulator, achieving 27.22 dB improvement in signal-to-noise and distortion ratio (SNDR) with 0.045 mW increase in power consumption. The proposed Σ-Δ DAC is designed with 180 nm CMOS technology with a supply voltage of 3.3 V. Benefiting from the optimization of the filtering mode and the precision trimming technique of the bandgap reference, an SNDR of 108.01 dB was achieved with a power consumption of 0.257 mW. In the temperature range of -40–105 °C, the temperature coefficient of the DAC output was less than 25.83 ppm/°C, under different process conditions.

1. Introduction

High-resolution, low-power digital-to-analog converters (DAC) are an inseparable part of applications in manufacturing, transportation, and telecommunications [1]. The size and matching requirements of typical Nyquist DACs increase significantly as the resolution increases. Therefore, the resolution is typically limited to <16-bit [2,3]. By contrast, Σ-Δ DACs can reduce element size and relax element matching requirements by using oversampling and noise shaping techniques. Therefore, Σ-Δ DACs are particularly suitable for high-precision applications.

As an essential component in the system, the Σ-Δ modulator can be classified into multibit [4,5] and single-bit [6] architectures based the number of quantization bits. For a modulator with a multibit architecture, the quantization noise can be reduced, and the analog filter circuit can be simplified. However, the number of DAC elements in this structure increases exponentially with the number of quantization bits, and the mismatch between the different elements affects the linearity of the DAC. Hence, it is necessary to shape the mismatch between the elements. Dynamic element matching (DEM) and data-weighted averaging (DWA) techniques can shape the element mismatch by cyclically selecting them; however, they significantly increase the complexity of circuit design and power consumption [7,8]. The peak signal-to-noise and distortion ratio (SNDR) of a modulator with single-bit architecture is relatively low. Its advantage is that it features a simple structure and high linearity without the utilization of a decoding circuit or mismatch-shaping circuit.

The performance of analog low-pass filter, which is used to filter out high-frequency noise of the Σ-Δ modulator, can significantly affect the effective number of bits (ENOB) of the entire DAC. As active filters, Chebyshev and Rauch filters [9,10] are utilized in voltage-mode Σ-Δ DACs to achieve sharp transition bands; however, considerable power is consumed by the operational amplifiers built in these active filters. Compared with the conventional switched-capacitor architecture in active filters, the power efficiency can be improved in a direct charge transfer (DCT) architecture [11]. However, active filters with a DCT architecture require switches to transfer charges. Therefore, the power consumption of the active filters increases as the sampling frequency increases.

To achieve a low-power Σ-Δ DAC, it is essential to reduce the power consumption of the analog filters. Benefiting from the current-mode architecture, high-frequency noise can be filtered out using a passive

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RC filter, which effectively reduces the power consumption of the Σ-Δ DAC. To guarantee the robustness of the Σ-Δ DAC, a bandgap reference was designed with precision trimming and high-order compensation techniques. The temperature coefficient (TC) of the DAC output current was less than 25.83 ppm/°C under different process conditions. A quantization noise randomization scheme was utilized within the Σ-Δ modulator, which delivered a 27.22 dB improvement in SNDR.

2. Architecture design of the proposed Σ-Δ DAC

The architecture of the proposed DAC is shown in Fig. 1. To reduce the power consumption of Σ-Δ DAC, the active filter was replaced with a current-mode low-pass configurable passive RC filter. The proposed DAC replaces the on-chip digital IF by providing oversampled digital codes ($f_{\mathrm{data}} = 1.024$ MHz). Since the input data is already at an interpolated rate, the Σ-Δ DAC can operate well without the on-chip digital IF. The circuit in the digital domain consists of a sampling-and-hold circuit, a Σ-Δ modulator, and a dither circuit for noise randomization. The analog part includes a 1-bit voltage-to-current (V/I) converter, a bandgap reference, and a configurable RC low-pass filter. With the 1-bit V/I converter, the code stream generated by the Σ-Δ modulator is converted to current. Owing to the high frequency noise in the output current of the V/I converter, the passive RC filter is designed to suppress the high frequency noise and provide a stable current for industrial transmitters. Compared with the Σ-Δ DAC with voltage-mode output, the DAC with current-mode output can provide stable current even with slight variations in load impedance. The bandgap reference was designed using a trimming and compensation technique to increase its robustness against PVT variations; thus, the TC drift of the DAC with process and voltage variations was low. Off-chip capacitors were utilized in the RC filter to reduce the complexity of on-chip circuits and increase the convenience of cutoff frequency adjustment. With the RC filter capacitance adjustment, the cutoff frequency can be adjusted to filter out noise of different frequency components. This adjustment can also regulate the settling time to match the update rate of the input digital code, under different operation conditions.

3. Circuit design

3.1. The Σ-Δ modulator

The structure of a traditional Σ-Δ modulator with 2-order is shown in Fig. 2 (a). The loop coefficients of the Σ-Δ modulator in Ref. [12] can be obtained by using Matlab. The signal transfer function (*STF*) and noise transfer function (*NTF*) of the Σ-Δ modulator can be obtained based on its OSR, architecture, order, and out-of-band gain. With this method, the generated coefficients of a modulator with an order of no less than two are typically achieved in a circuit using multipliers, adders, and shift registers, which significantly increase the power consumption of the modulator.

To reduce the power consumption of the Σ-Δ modulator, a multiplier-free architecture was used in this study, as shown in Fig. 2 (b). Multiple loop coefficients are necessary to form the traditional

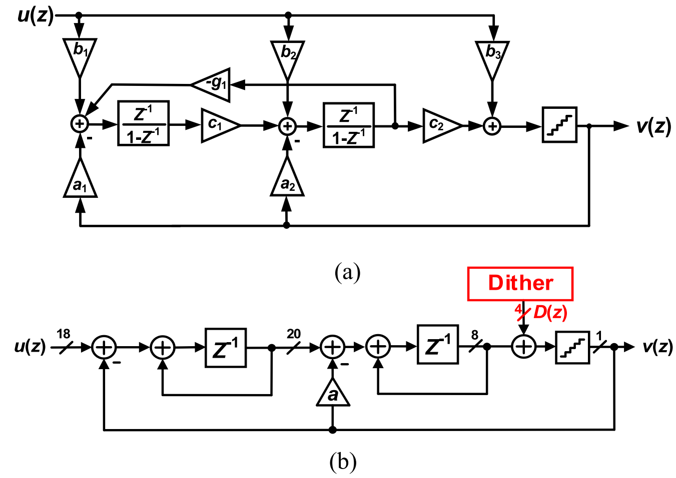


Fig. 2. Σ-Δ modulators with (a) traditional and (b) multiplier-free architecture.

architecture. By contrast, only one feedback coefficient exists in the proposed architecture, which significantly simplifies the circuit complexity and substantially reduces the power consumption of the modulator.

For a Σ-Δ DAC with resolution of over 18-bit, a peak SNDR of over 120 dB is required for its modulator. The SNDR variations of the Σ-Δ modulators with different orders versus OSR are shown in Fig. 3. For a 1-order Σ-Δ modulator, even with a high OSR of 4096, the peak SNDR of the modulator is still less than 120 dB, and the high OSR can significantly increase the power consumption of other circuits shown in Fig. 1. For a 3-order Σ-Δ modulator without loop coefficients, overflow occurs in the modulator, which limits its dynamic range. As shown in Fig. 3, the traditional 3-order Σ-Δ modulator can achieve a peak SNDR of over 120 dB with an OSR of 256; however, the complicated circuit occupies considerable area. Benefiting from a simple multiplier-free structure, the power consumption of the proposed Σ-Δ modulator with 2-order structure is reduced by 54.6 %, compared with the traditional 2-order modulator. Based on the above comparison, to achieve a peak SNDR of over 120 dB and reduce the power budget of the DAC, the proposed Σ-Δ modulator was designed with 2-order and an OSR of 2048.

The transfer function of the Σ-Δ modulator shown in Fig. 2 (b) can be obtained using Eq. (1), where $Q(z)$ and $D(z)$ are the noise of the quantizer and dither, respectively. $STF(z)$ and $NTF(z)$ are given by Eq. (2).

$$v(z) = STF(z)u(z) + NTF(z)[Q(z) + D(z)] \tag{1}$$

$$\begin{cases} STF(z) = \dfrac{z^{-2}}{1 + (a-2)z^{-1} + (2-a)z^{-2}} \\ NTF(z) = \dfrac{(1-z^{-1})^2}{1 + (a-2)z^{-1} + (2-a)z^{-2}} \end{cases} \tag{2}$$

According to Eq. (2), the pole-zero locations of the *NTF* can be obtained and shown in Fig. 4 (a), where the two poles located inside the unit circle guarantee the stability of the 2-order Σ-Δ modulator with

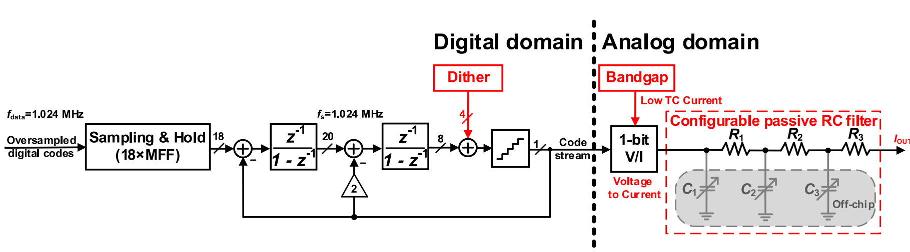


Fig. 1. Architecture of the proposed Σ-Δ DAC.

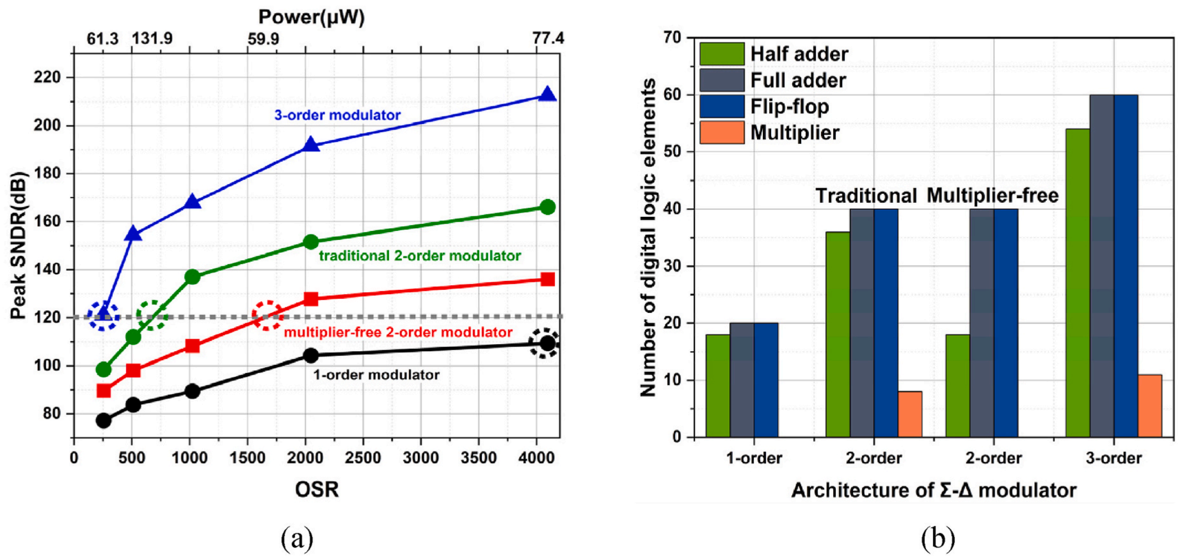


Fig. 3. Different orders of Σ-Δ modulators with (a) SNDR and (b) number of digital logic elements.

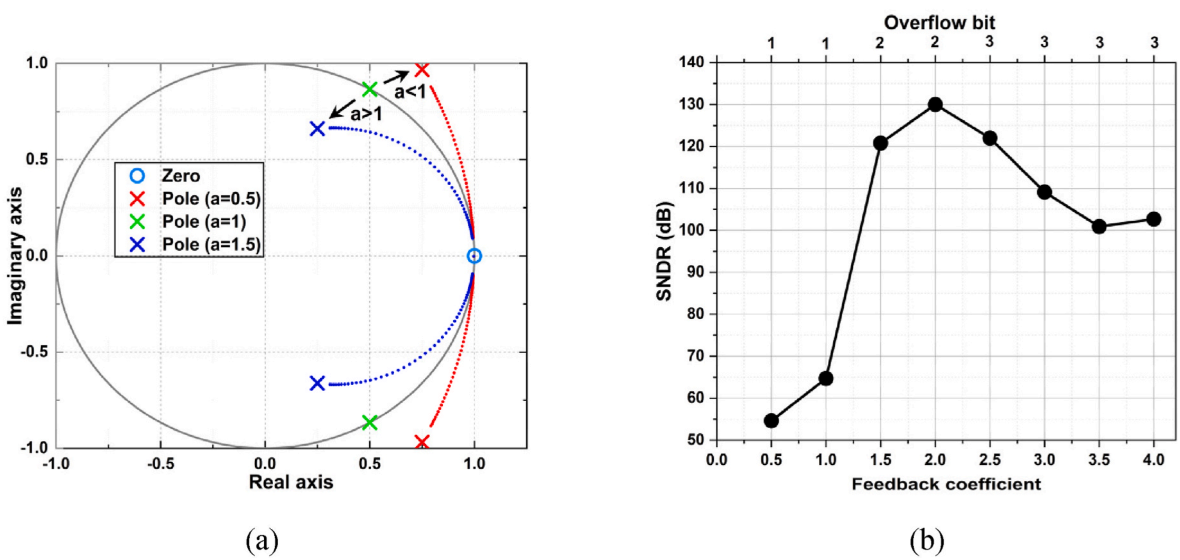


Fig. 4. Different feedback coefficients for (a) pole-zero locations of the *NTF* and (b) SNDR of the modulator.

a>1. The SNDR of the Σ-Δ modulator versus the feedback coefficient is shown in Fig. 4 (b); the Σ-Δ modulator achieves the maximum SNDR with only two overflow bits at a feedback coefficient of 2. Based on the above comparison, the feedback coefficient of the proposed Σ-Δ modulator was set to 2.

The integrator was implemented using half adders, full adders, and MFFs, as shown in Fig. 5. First, the accumulation of the input signal *DIN*<17:0> and feedback signal *Z* is achieved using half adders. Subsequently, the outputs of the half adders and the MFFs are accumulated using full adders. Finally, the output of the full adders is delayed by the MFFs and sent to the next integrator. In Fig. 5, *HCO*<16:0> is the 17-bit carry output of the half adder, *HS* is the sum output of the half adder, *FCO* <18:0> is the 19-bit carry output of the full adder, *CLK* is the clock

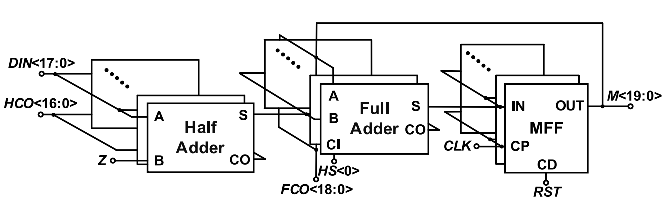


Fig. 5. Structure of the integrator.

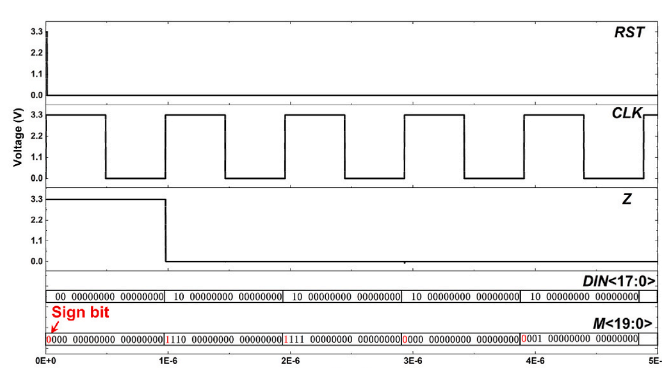


Fig. 6. Operation of the integrator.

signal, and RST is the reset signal.

The operation of the integrator is shown in Fig. 6. With the rising edge of CLK, the integrator achieves the function of addition or subtraction based on the input signal DIN<17:0> and feedback result Z, where the most significant bit (MSB) of the integrator output, M<19>, is the sign bit.

3.2. Dither circuit

To reduce the limitation of idle tones on the SNDR of the Σ-Δ modulator, dither is utilized to randomize the quantization noise. Resistors and noise diodes are typically used to generate true random signals in true-random signal generation circuits [13]. However, the true random signal generated by this architecture has low amplitude. Operational amplifiers are typically used to amplify the true random signal [14]. In pseudo-random signal generation circuits, a linear feedback shift register (LFSR) is typically used to generate pseudo-random signals [15]. The pseudo-random digital signal generated by the LFSR can directly match the input of the 1-bit V/I converter, thereby eliminating the need to use operational amplifiers [16].

Dither circuits of the conventional LFSR architecture are with a single feedback structure. Pseudo-random sequences of these circuits are characterized by simple generation logic and strong correlation [17]. The proposed dither circuit varies the generation logic of pseudo-random sequences by adding feedback paths, thereby improving the randomness among sequences. To guarantee the length and randomness of the pseudo-random sequences while reducing the power consumption and area of the dither circuit, 15 flip-flops and 4 XNOR gates were utilized in the proposed dither circuit. The structure of the dither circuit in this study is a 15-bit linear feedback shift register (LFSR), as shown in Fig. 7.

The generation logic of the pseudo-random sequences D_1, D_2, D_3, and D_4 is shown in Eq. (3). To generate the sequences for the current period, the dither circuit performs XNOR operation on the sequences of the previous periods.

$$\begin{cases} D_1 = Z^{-4}\{D_4 \odot [Z^{-3}\cdot(D_1 \odot D_2)]\} \\ D_2 = Z^{-4}(D_1 \odot D_3) \\ D_3 = Z^{-3}(D_2 \odot D_4) \\ D_4 = Z^{-4}(D_1 \odot D_2) \end{cases} \tag{3}$$

At a sampling rate f_s of 1.024 MHz, the output sequence length of this circuit is $(2^{15}\text{-}1)/f_s = 31.999$ ms, as shown in Fig. 8. The pseudo-random sequences D_1, D_2, D_3, and D_4 generated by the dither circuit are shown in Fig. 9.

Owing to the randomization of the quantization noise, the pseudo-random sequences reduce the quantization noise within the 250 Hz bandwidth, leading to a 3.5 dB improvement in the SNDR of the DAC. However, additional noise is also introduced by the pseudo-random sequences. This noise is shaped to high frequency by the Σ-Δ modulator. With the introduction of the dither circuit, the high frequency noise in the spectrum increases.

Fig. 10 shows the output spectrum of the proposed Σ-Δ modulator. The update rate of the input data is 1.024 MHz. At a sampling frequency of 1.024 MHz and an input signal frequency of 187.5 Hz, the SNDR of the

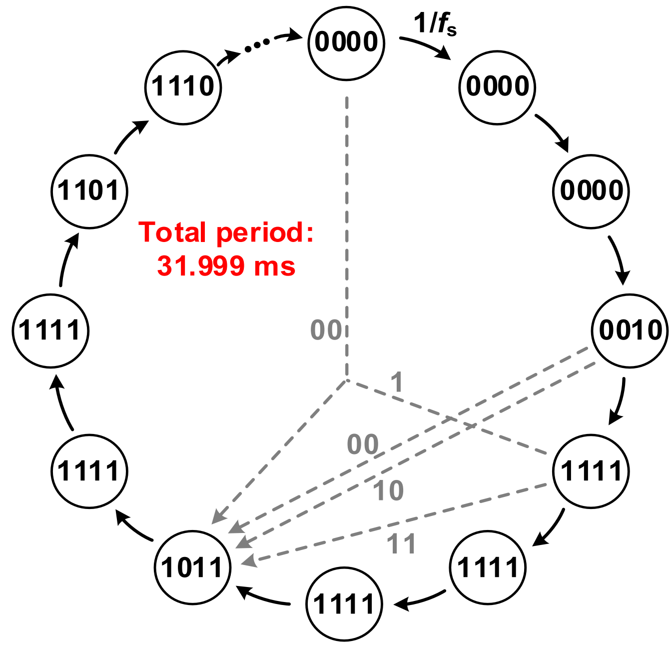


Fig. 8. State transfer of the dither circuit.

modulator without the dither was 92.82 dB. Owing to the noise randomization function introduced by the dither circuit, the SNDR of the modulator was improved to 120.04 dB.

3.3. Low-temperature-drift bandgap reference

Fig. 11 shows the bandgap reference circuit used in the proposed DAC. To increase the robustness of the operational amplifier-free bandgap reference against PVT variations, high-order temperature compensation and current-trimming techniques were employed.

In Fig. 11, the output current of the bandgap reference can be expressed as

$$I_{OUT} = x\cdot\left(\frac{V_{BE2} + (I_1 + I_2 + I_3 + I_{DAC})\cdot R_2}{R_5} + I_6 + I_7\right) \tag{4}$$

where V_{BE2} is the base-emitter voltage of transistor Q_2 and can be expressed as [18].

$$V_{BE} = V_G(T_0) + \frac{T}{T_0}[V_{BE}(T_0) - V_G(T_0)] + (\eta - \alpha)\frac{kT}{q}\ln\frac{T}{T_0} \tag{5}$$

where V_G is the bandgap voltage of silicon extrapolated at 0 K, T is the absolute temperature in degrees kelvin, η is a temperature constant depending on technology, α is the order of the temperature dependence of the collector current, and T_0 is the reference temperature. According to Ref. [19], currents I_1, I_2, and I_3 can be expressed as

$$I_1 = I_2 = \frac{V_{BE2} - V_{BE1}}{R_1} = \frac{kT}{q}\cdot\frac{\ln(n)}{R_1} \tag{6}$$

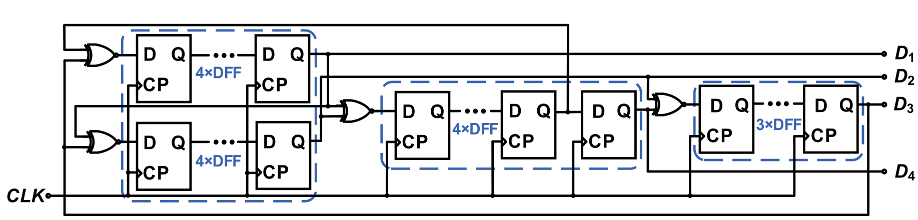


Fig. 7. Structure of the dither circuit used in the proposed DAC.

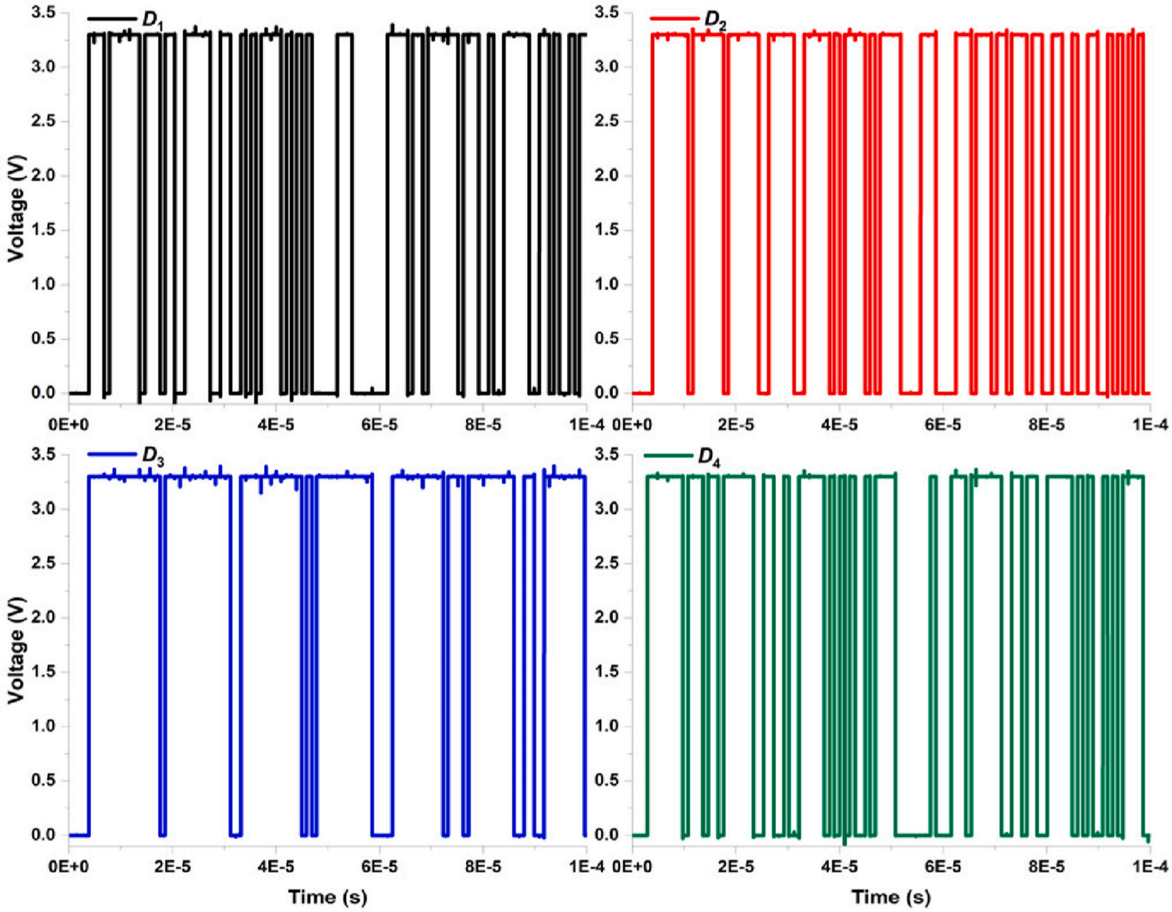


Fig. 9. Pseudo-random sequences of the dither.

$$I_3 = \frac{V_{BE2} - V_{BE3}}{R_3} = \frac{V_T \cdot \ln(m)}{R_3} = -\frac{kT}{q} \cdot \ln\frac{T}{T_0} \tag{7}$$

where n is the emitter area ratio of Q_1 and Q_2, and m is the emitter area ratio of Q_3 and Q_2. Combining Eqs. (1)–(4), V_{OUT} can be expressed as

$$I_{OUT} = x \cdot \left\{ \left[V_G(T_0) + \frac{T}{T_0} \cdot [V_{BE2}(T_0) - V_G(T_0)] + 2 \cdot \frac{kT}{q} \cdot \frac{R_2}{R_1} \cdot \ln(n) + I_{DAC} \cdot R_2 + (\eta - R_2 - 1) \cdot \ln\frac{T}{T_0} \cdot \frac{kT}{q} \right] \cdot \frac{1}{R_5} + I_6 + I_7 \right\} \tag{8}$$

The nonlinear term $(\eta\text{-}R_2\text{-}1)\bullet\ln(T/T_0)\bullet kT/q$ can be eliminated by adjusting resistance R_2. The compensation current I_{DAC} compensates for the reference current when the process varies. With this compensation, the temperature coefficient of the reference current could be significantly decreased. To guarantee the trimming performance and reduce the complexity of the trimming circuit, an 8-bit trimming DAC was designed, as shown in Fig. 11. I_{DAC} has a positive temperature coefficient. By controlling DN<7:0>, the output current I_{DAC} can be varied.

Fig. 12 shows the variation in the bandgap reference output with temperature. The output current before trimming varies significantly in the temperature range of -40–105 °C. However, in the worst case, the variation in the output current after trimming was less than 41.13 nA, and the corresponding temperature coefficient of the output current was 25.83 ppm/°C.

4. Design results and comparison

The proposed 18-bit Σ-Δ DAC is designed with 180 nm CMOS technology. The layout of the proposed DAC is shown in Fig. 13. The area is 385.27 μm × 497.17 μm; the area breakdown is shown in Fig. 14. The

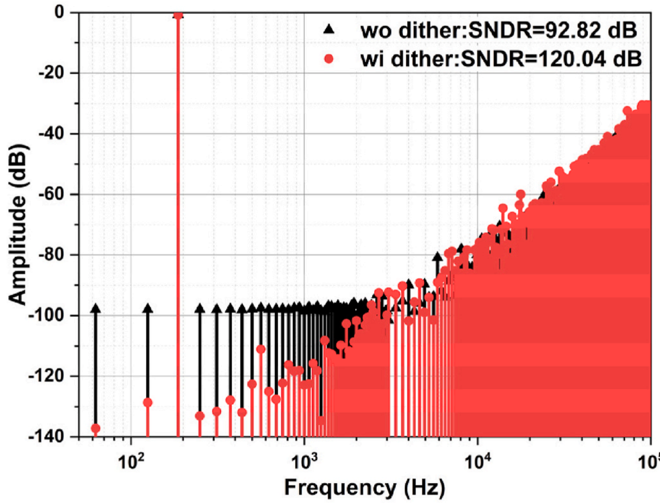


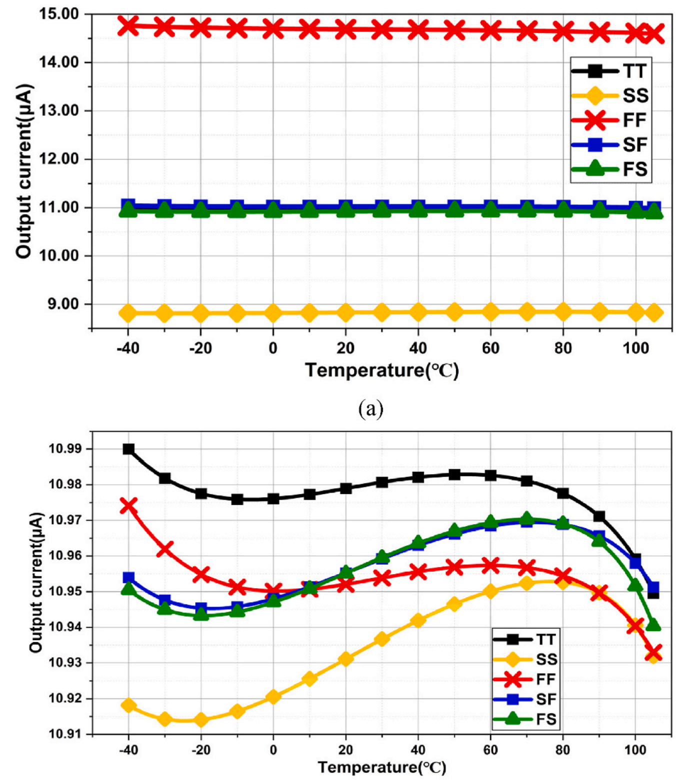
Fig. 10. Output spectrum of the proposed Σ-Δ modulator.

bandgap reference and dither circuits occupy 54.93 % and 8.97 % of the total area, respectively. The sampling-and-hold (S&H) circuit, modulator, and V/I converter occupy 5.68, 24.15, and 6.27 % of the total area, respectively, which is equivalent to 36 % of the total area of the core circuit of the proposed DAC.

Fig. 15 shows the power consumption of the proposed Σ-Δ DAC. The design results showed that the power consumption of the DAC was 0.257 mW at a sampling rate of 1.024 MHz and supply voltage of 3.3 V. The bandgap reference and dither circuits accounted for 19.81 % and 17.68 % of the total power consumption, respectively. The sampling-and-hold circuit, modulator, and V/I converter accounted for 6.76, 49.98, and 5.76 % of the total power consumption, respectively, which is equivalent to 62.5 % of the total power consumed by the entire DAC circuit.

Fig. 16 shows the output spectrum of the proposed Σ-Δ DAC. The dynamic performance of the Σ-Δ DAC was analyzed with a supply voltage of 3.3 V, sampling rate and input data update rate of 1.024 MHz, and signal frequency of 187.5 Hz. The resistors R_1, R_2, and R_3 in the RC filter are 15 kΩ, and the off-chip capacitors C_1, C_2, and C_3 are 2.2 nF. The harmonics are partially attenuated by the RC filter. Based on the application of industrial transmitters, the bandwidth for the FFT analysis was set to 250 Hz. Benefiting from the quantization noise randomization technique, the SNDR of the Σ-Δ DAC improved by 3.5 dB. The spurious-free dynamic range (SFDR), SNDR, and ENOB of the proposed Σ-Δ DAC were 111.34 dB, 108.01 dB, and 17.65 bits, respectively. The SNDR of the Σ-Δ DAC under different PVT conditions is given in Fig. 17.

The performance of the proposed Σ-Δ DAC and comparison with the state-of-the-art DACs are summarized in Table 1. The mismatch of DAC elements in Ref. [20] is effectively shaped by combining vector element selection with fixed transitioning of elements. Although this DAC



(a)

(b)

Fig. 12. Variation in bandgap reference output with temperature (a) before and (b) after trimming.

achieved a superior SNDR, the mismatch shaping circuit significantly increased its power consumption. Time-interleaving technique was employed to widen the bandwidth of the Σ-Δ DAC in Ref. [22]. An inverse-sinc-shaped digital predistortion (DPD) scheme was used to improve the linearity of the DAC. However, a signal generator was used to bias the reference voltage in the DAC. The effect of the bandgap reference on the DAC was not considered in Ref. [22].

In this study, a bandgap reference with high-order compensation and precision trimming techniques were employed in designing the proposed Σ-Δ DAC. The temperature coefficient of the Σ-Δ DAC output was less than 25.83 ppm/°C under different process conditions, which guarantees the robustness of the DAC. The proposed DAC is designed with 180 nm CMOS technology. Owing to the multiplier-free modulator and the off-chip passive RC filter, the power and area consumption disadvantages from the perspective of process advancement can be effectively mitigated. The proposed Σ-Δ DAC is competitive in power consumption.

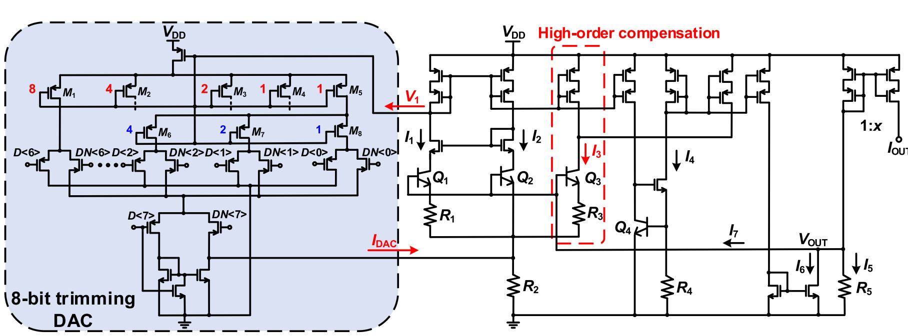


Fig. 11. Bandgap reference used in the proposed DAC.

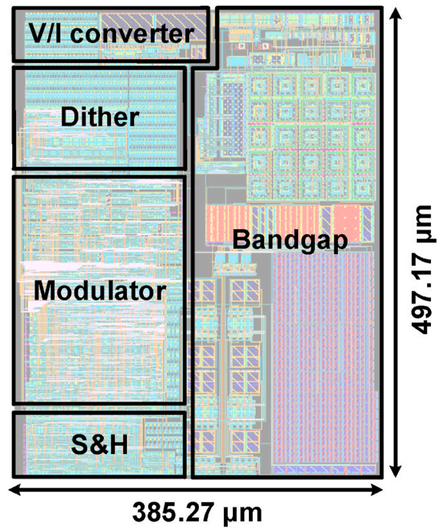


Fig. 13. Layout of the proposed Σ-Δ DAC.

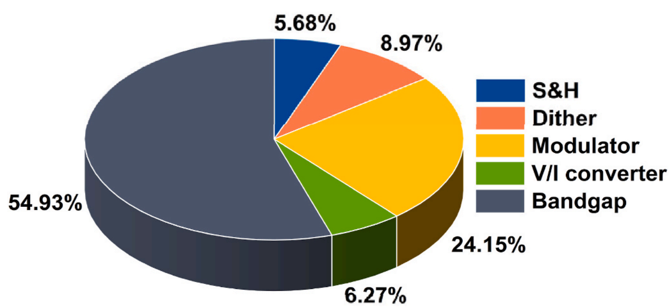


Fig. 14. Area breakdown of the proposed Σ-Δ DAC.

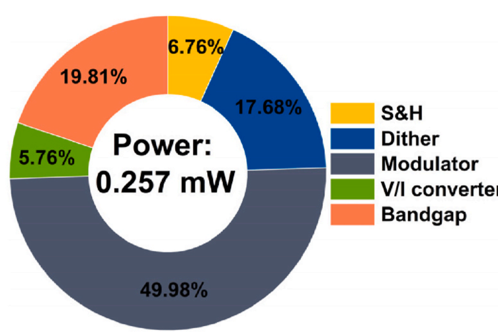


Fig. 15. Power consumption breakdown of the proposed Σ-Δ DAC.

5. Conclusion

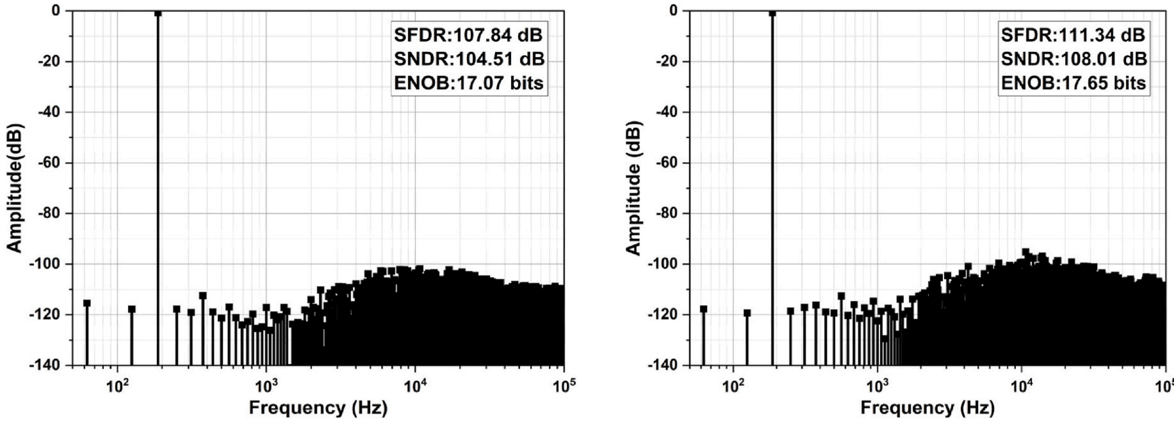
An 18-bit low-power Σ-Δ DAC with low-temperature-drift reference was proposed. To reduce the power consumption of the DAC, the active filter was replaced with a current-mode low-pass configurable passive RC filter. The capacitors of the RC filter were designed off-chip to reduce the complexity of the on-chip circuits and increase the convenience of cutoff frequency adjustment. Owing to the use of high-order compensation and trimming techniques, the bandgap reference provided strong robustness against PVT variations, which effectively guarantee the robustness of the proposed Σ-Δ DAC. A dither circuit for noise randomization provided a 3.5 dB improvement in SNDR. Designed with a 180 nm CMOS technology, the proposed Σ-Δ DAC occupied an active area of 0.191 mm^2. At a sampling rate and input data update rate of 1.024 MHz, supply voltage of 3.3 V, bandwidth of 250 Hz, and signal frequency of 187.5 Hz, the proposed Σ-Δ DAC achieved an ENOB of 17.65 bits while consuming only 0.257 mW of power. The proposed Σ-Δ DAC exhibits competitive accuracy and PVT robustness with low power consumption, making it suitable for industrial transmitter applications.

CRediT authorship contribution statement

Xingyuan Tong: Writing – review & editing, Investigation. **Hao Yu:** Writing – review & editing. **Xin Xin:** Writing – review & editing. **Yuhua Liang:** Writing – review & editing, Funding acquisition.

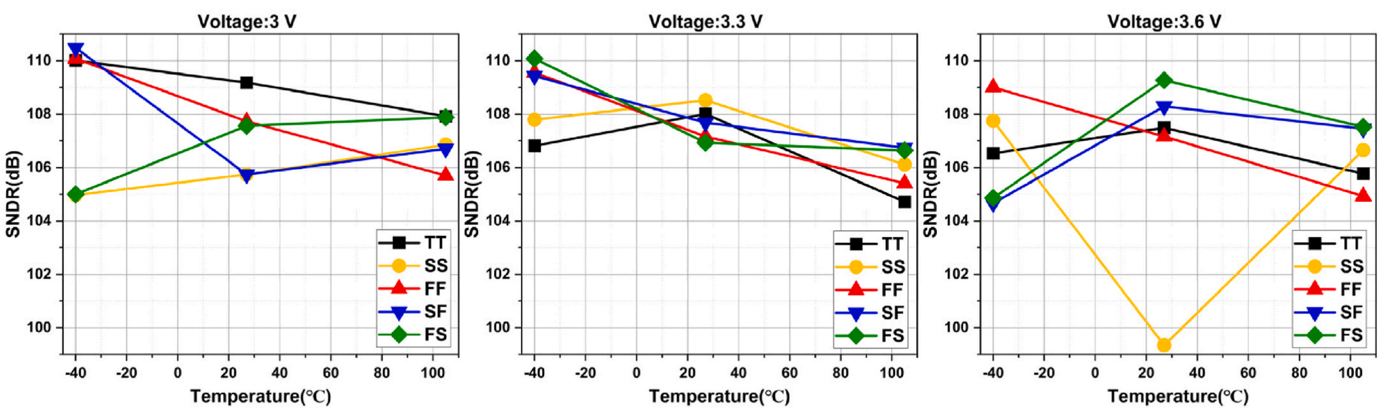
Data availability statement

Date sharing is not applicable to this article as no datasets were generated or analyzed during the current study.



(a) (b)

Fig. 16. Output spectrum of the proposed Σ-Δ DAC (a) without and (b) with dither.



(a) (b) (c)

Fig. 17. Design results of the proposed Σ-Δ DAC under different PVT conditions.

Table 1
Performance summary and comparison with state-of-the-art DACs.

	TCAS-II '2022 [2]	CICC '2019 [20]	TCAS-II '2024 [21]	JSSC '2018 [22]	VLSI '2021 [23]	TCAS-II '2024 [24]	Proposed[a]
Resolution (bit)	14	24	10	16	16	16	18
Technology (nm)	40	40	65	65	65	40	180
Supply (V)	1/2.5	1.15/8	2.5	1/2.5	3.3	1/1.8	3.3
Bandwidth (Hz)	$375{\cdot}10^6$	$20{\cdot}10^3$	$268{\cdot}10^6$	$2.25{\cdot}10^9$	–	$43{\cdot}10^6$	250
Modulator order	–	2	2	3	2	–	2
Quantization (bit)	–	6	1	4	1	–	1
Filter type	–	active	active	active	passive	active	passive
SNDR (dB)	–	135	30.74	–	74	–	108.01
SFDR (dB)	63	–	49.38	66	–	80.8	111.34
Power (mW)	190	143	263.11	1140	–	360	0.257
FoM_1 (dB)	–	186.45	120.82	–	–	–	167.89
FoM_2 (dB)	155.95	–	139.46	158.95	–	161.57	171.22
Temperature-insensitive bias	No	Yes	Yes	No	No	No	Yes

[a] Simulation results; FoM_1 = SNDR+10lg(BW/Power); FoM_2 = SFDR+10lg(BW/Power).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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